

Retrained pixel classifier on 2x2 image tiles - Hackathon-Polaron BSE

Answer: the model reaches **98.9%** accuracy on the held-out test images (95% CI [98.4, 99.4], bootstrapped over 7 whole test images; balanced accuracy 97.0% [95.6-98.1]). The 95% target is **REACHED** at the CI lower bound. **Important caveat:** the labels were produced by the 3-class multi-Otsu pipeline (qc_real.py), not by a person - so these numbers measure *agreement with that thresholding method*, not correctness against ground truth. Bright particles are assumed silicon-based and the grey bulk graphite, from appearance only.

Method

Each of the 31 BSE images was cut into a 2x2 grid (124 tiles, ~3500x1000 px each); Inlens and ETD were cut identically to stay aligned. Labels are the multi-Otsu masks computed once per full image and cut the same way. A RandomForestClassifier (60 trees, max depth 32) predicts the label of each pixel from 12 named features (smoothed intensity at three scales, relative brightness, gradient, two local-std textures, local min/max, and the smoothed Inlens and ETD channels). Training used a random subsample of 30k pixels per tile with geometric augmentation (h/v flips, rot180) on training tiles only. Splitting is by *image*: all four tiles of an image stay together; 7 images (2/2/3 per batch) are held out as a final test set used once; the remaining 24 go through 5-fold grouped cross-validation.

Results

CV (correct, grouped by image): accuracy 0.9774 +/- 0.0154 across 5 folds; balanced 0.9573 +/- 0.0130.

class	precision	recall	F1	IoU
pore	0.990	0.976	0.983	0.967
bulk	0.989	0.997	0.993	0.987
bright	0.984	0.920	0.951	0.906

Phase-fraction error vs labels (the downstream use): mean abs error 0.0048 [0.0024-0.0077] of image area per class. Majority-class baseline = 0.816; plain per-tile multi-Otsu baseline agreement = 0.977.

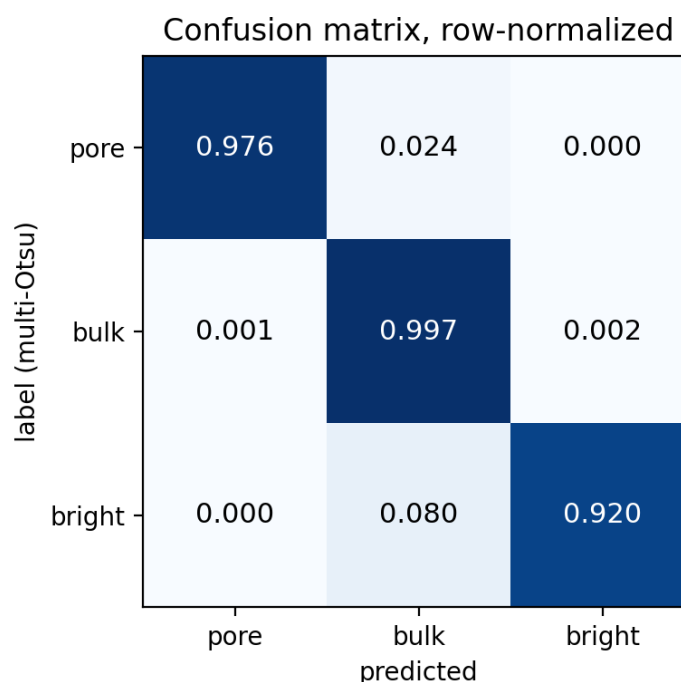


Fig 1. Row-normalized confusion on held-out test pixels.

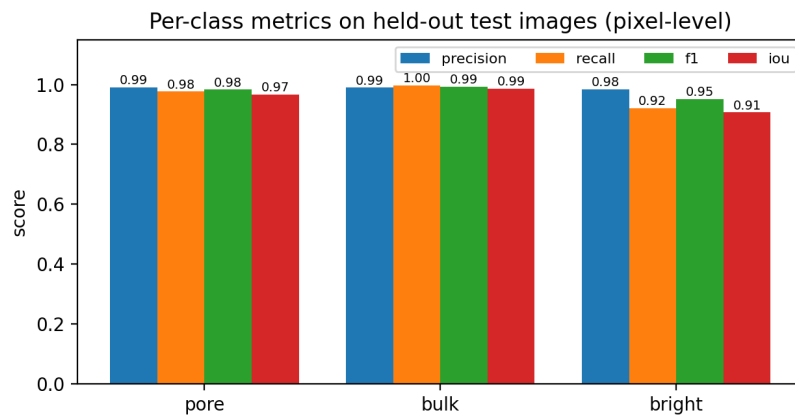


Fig 2. Per-class precision/recall/F1/IoU on test.

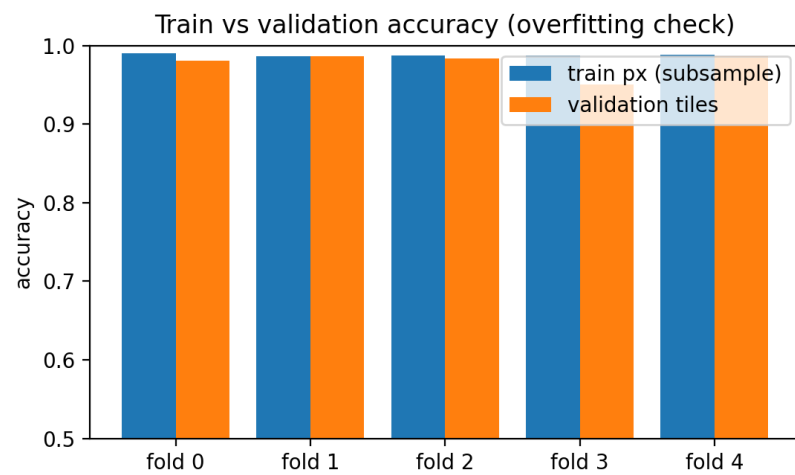


Fig 3. Train vs validation accuracy per fold.

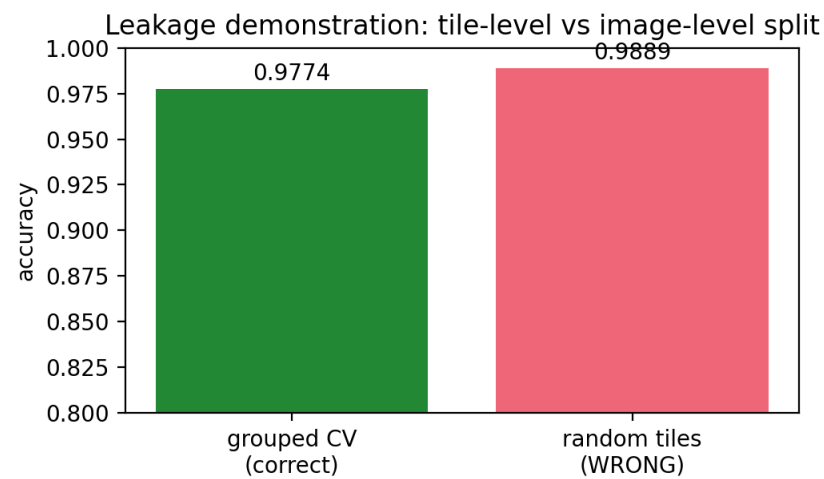


Fig 4. Random tile split (wrong) vs grouped split.

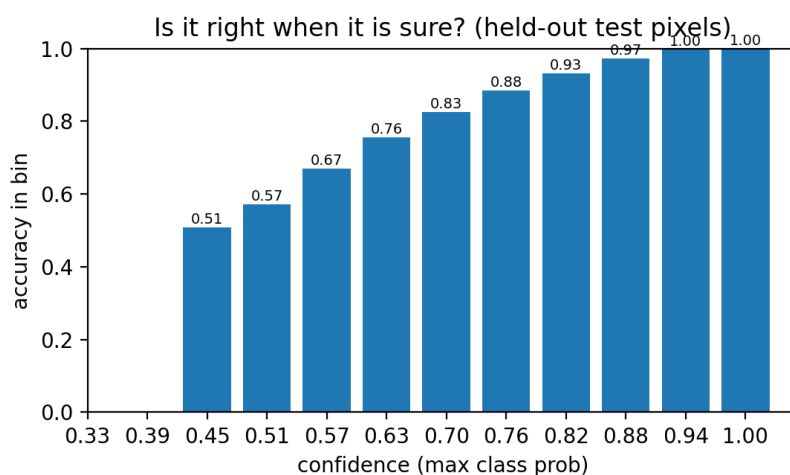


Fig 5. Accuracy by confidence bin on test pixels.

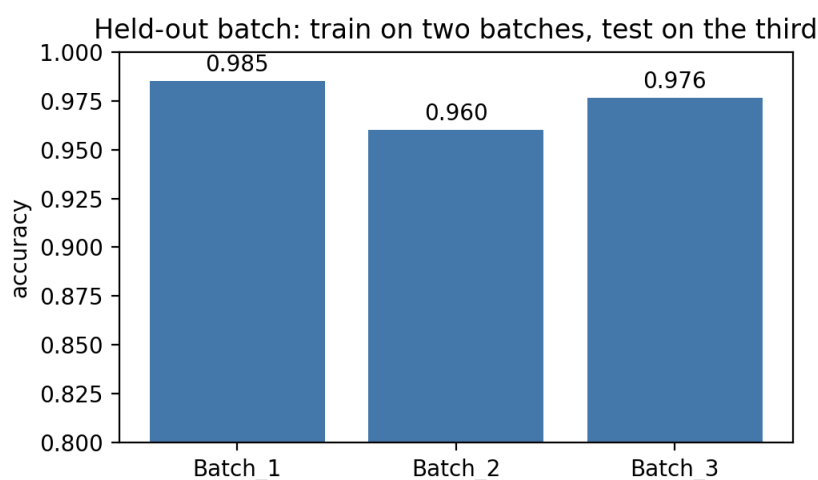


Fig 6. Train on two batches, test on the third.

Sanity checks

1_leakage_assert: PASS; 2_leakage_demonstration_WRONG: PASS; 3_shuffled_labels: PASS;
 4_baselines: PASS; 5_overfit_train_vs_val: PASS; 6_tile_edge: PASS; 7_reassembly: PASS;
 8_reproducibility: PASS; 9_heldout_batch: PASS; 10_eyeball_examples: PASS

Known failure cases and limitations

Accuracy = agreement with multi-Otsu, not human truth. Held-out-batch accuracies: Batch_1 0.985; Batch_2 0.960; Batch_3 0.976. Worst tiles are listed in results.xlsx (Worst_tiles) and shown in figures/worst_tiles.png; errors concentrate at class borders, dark grey regions read as pore by the labeler, and low-contrast outlier images. Confusion is mostly pore<->bulk at soft edges; see results.xlsx Confusion_test. Model and labels are 2D image quantities; pixel size ~25 nm/px. The two outlier Batch_1 images remain in the data and are flagged in Per_image_test.